VICTORY 2027

Climate

LECTURE-3 Geography

Sameer Sir



Topics To Be Covered

1. Monsoons and its mechanism (Part 02)
2. Seasons
3.
4.

Climate

Distribution of Temperature, Rainfall, winds in Summer and Winter and factors affecting the climate of India (latitudes, altitude, distance from sea, winds, Himalayas, jet streams, and El Nino). Monsoon and its mechanism. Seasons: Summer; Monsoon, Retreating Monsoon and Winter. Temperature and pressure conditions, wind movements, local winds, precipitation and interpretation of climate data.

Continentality → A large area surrounded by land that gets heated up and cooled quickly.

↳ Extreme temperatures

Too Hot / Too Cold.

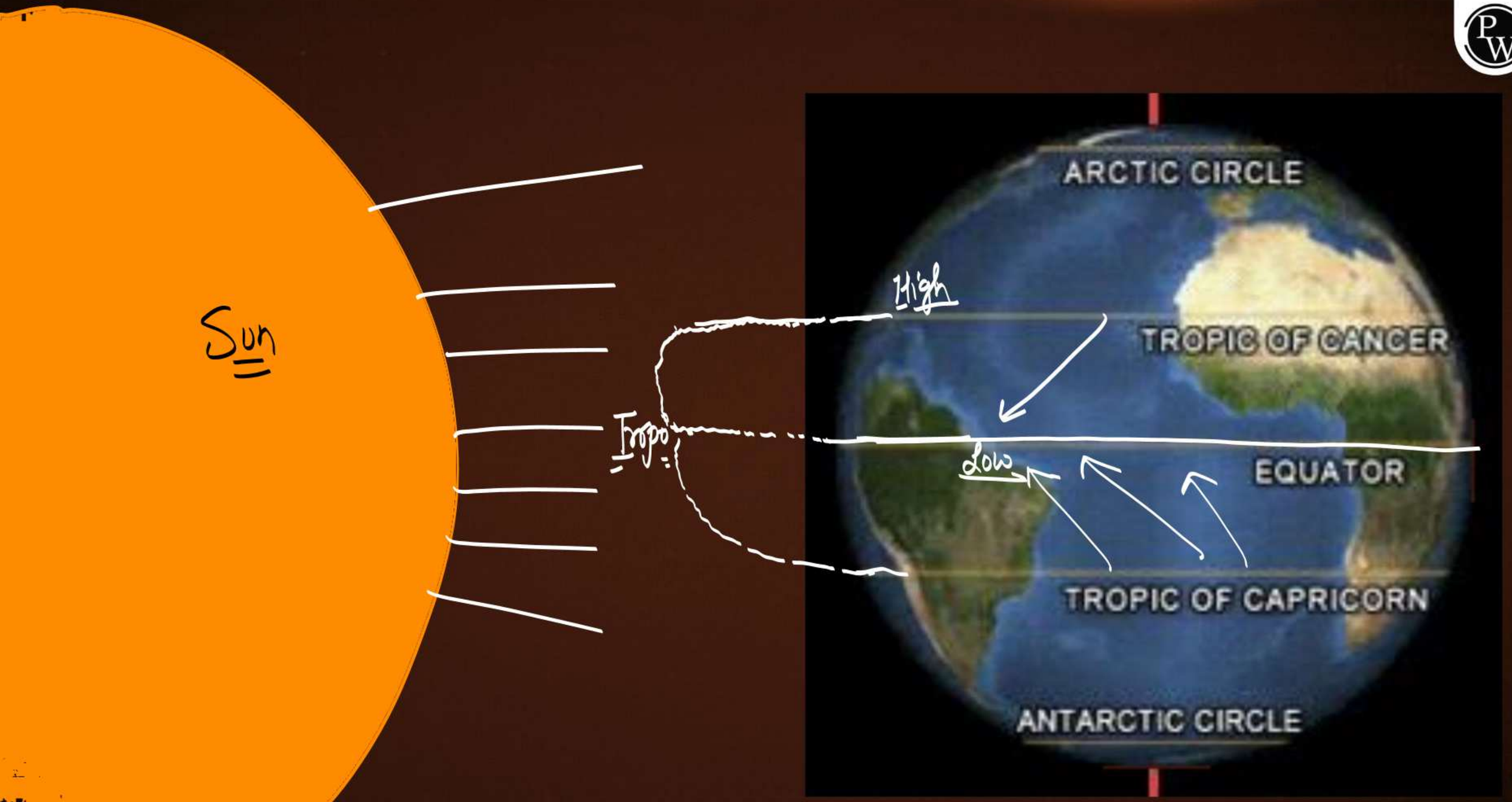
Maritime influence → Surrounded by sea. Moderate temperature.

Phenomena of Monsoon → South West Monsoon Winds. } Onset of Monsoon

North East Monsoon } Retreating monsoon


ARCTIC OCEAN
Greenland
(Denmark)
Svalbard
(Norway)
ARCTIC OCEAN
United States
Canada
Iceland
Norway
Sweden
Finland
Russia
U.K.
Ireland
Estonia
Latvia
Lithuania
Belar.
Ukraine
Mol.
Poland
Germany
France
Spain
Portugal
Italy
Rom.
Bul.
Greece
Turkey
Cyprus
Armenia
Geor.
Az
Kazakhstan
Mongolia
Uzbek.
Turkm.
Kyrgyzstan
Tajikistan
China
Iran
Iraq
Syr.
Lebanon
Israel
Jordan
Kuwait
Qatar
Saudi Arabia
U.A.E.
Oman
Yemen
Afghan.
Pakistan
India
Nepal
Bhutan
Bangl.
Bur.
Laos
Thai.
Vietnam
Cambodia
Taiwan
Philippines
N. Korea
S. Korea
Japan
Summer
United States
Mexico
Caribbean
Central America
Guyana
Suriname
French Guiana (France)
Venez.
Colombia
Ecuador
Peru
Brazil
Bolivia
Paraguay
Chile
Argentina
Uruguay
Morocco
Tunisia
Algeria
Libya
Egypt
Western Sahara
Senegal
Maurit.
Mali
Niger
Chad
Sudan
Eritrea
Djibouti
The Gambia
Guinea-Bissau
Guinea
Sierra Leone
Liberia
Ghana
Togo
Benin
Cote d'Ivoire
Nigeria
Cam.
C.A.R.
South Sudan
Ethiopia
Somalia
Ken.
Uganda
Rwanda
Burundi
Dem. Rep. of Congo
Congo
Gabon
Equatorial Guinea
Tanz.
Angola
Zambia
Zim.
Namibia
Bots.
South Africa
Lesotho
eSwatini
Malawi
Mozambique
Madagascar
Sri Lanka
Malaysia
Indonesia
Timor-Leste
Papua New Guinea
Solomon Islands
New Caledonia
Fiji
Australia
New Zealand
PACIFIC OCEAN
PACIFIC OCEAN
ATLANTIC OCEAN
ATLANTIC OCEAN
INDIAN OCEAN
SOUTHERN OCEAN
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Sun
High
Trop
Low
ARCTIC CIRCLE
TROPIC OF CANCER
EQUATOR
TROPIC OF CAPRICORN
ANTARCTIC CIRCLE


INDIA
States and Union Territories
INDIA FLAG
TAJIKISTAN
AFGHANISTAN
PAKISTAN
CHINA (TIBET)
NEPAL
BHUTAN
BANGLADESH
MYANMAR (BURMA)
SRI LANKA
LADAKH
Srinagar
Leh
JAMMU & KASHMIR
Jammu
HIMACHAL PRADESH
Shimla
PUNJAB
CHANDIGARH
Dehradun
UTTARAKHAND
HARYANA
DELHI
New Delhi
Jaipur
RAJASTHAN
UTTAR PRADESH
Lucknow
SIKKIM
Gangtok
ARUNACHAL PRADESH
Itanagar
Dispur
ASSAM
NAGALAND
Kohima
Shillong
MEGHALAYA
Imphal
MANIPUR
Agartala
TRIPURA
Aizawl
MIZORAM
BIHAR
Patna
JHARKHAND
Ranchi
WEST BENGAL
Kolkata
Gandhinagar
GUJARAT
Bhopal
MADHYA PRADESH
CHHATTISGARH
Raipur
ODISHA
Bhubaneswar
BAY OF BENGAL
Daman
DADRA & NAGAR HAVELI AND DAMAN & DIU
MAHARASHTRA
Mumbai
TELANGANA
Hyderabad
Amaravati
Yanam (Puducherry)
ARABIAN SEA
Panaji
GOA
ANDHRA PRADESH
KARNATAKA
Bengaluru
Chennai
Mahe (Puducherry)
PUDUCHERRY
LAKSHADWEEP (INDIA)
KERALA
Kavaratti
TAMIL NADU
Karaikal (Puducherry)
Palk Strait
Thiruvananthapuram
ANDAMAN & NICOBAR ISLANDS (INDIA)
Port Blair
International Boundary
State/UT Boundary
Country Capital
State/UT Capital
Map not to Scale
INDIAN OCEAN
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Himalayas
(The Bouncer)

Latitude
(The Divider)



Altitude
(The Cooler)

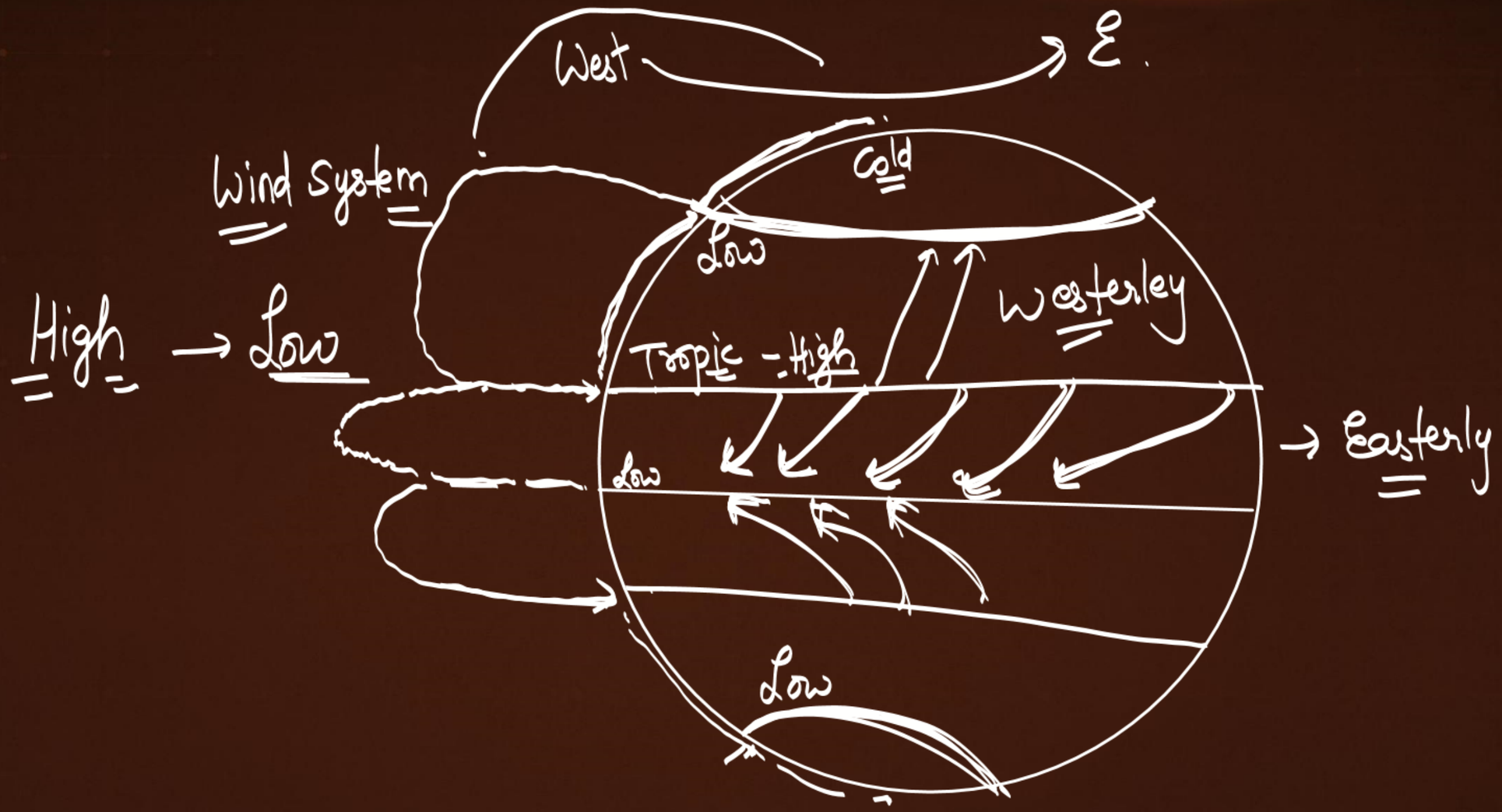
Relief
(The Squeezer)

Summer
SW Monsoon
(The Inflow)

Winter
NE Retreat
(The Outflow)

India doesn't just have one climate; it has an intricate, interconnected ecosystem. Extreme altitudes, mountain shields, and shifting oceanic winds combine every single day to create the most dynamic weather puzzle on Earth.

NotebookLM


West → E.
Wind System
High → Low
Cold
Low
Westerley
Tropic - High
→ Easterly
Low
Low

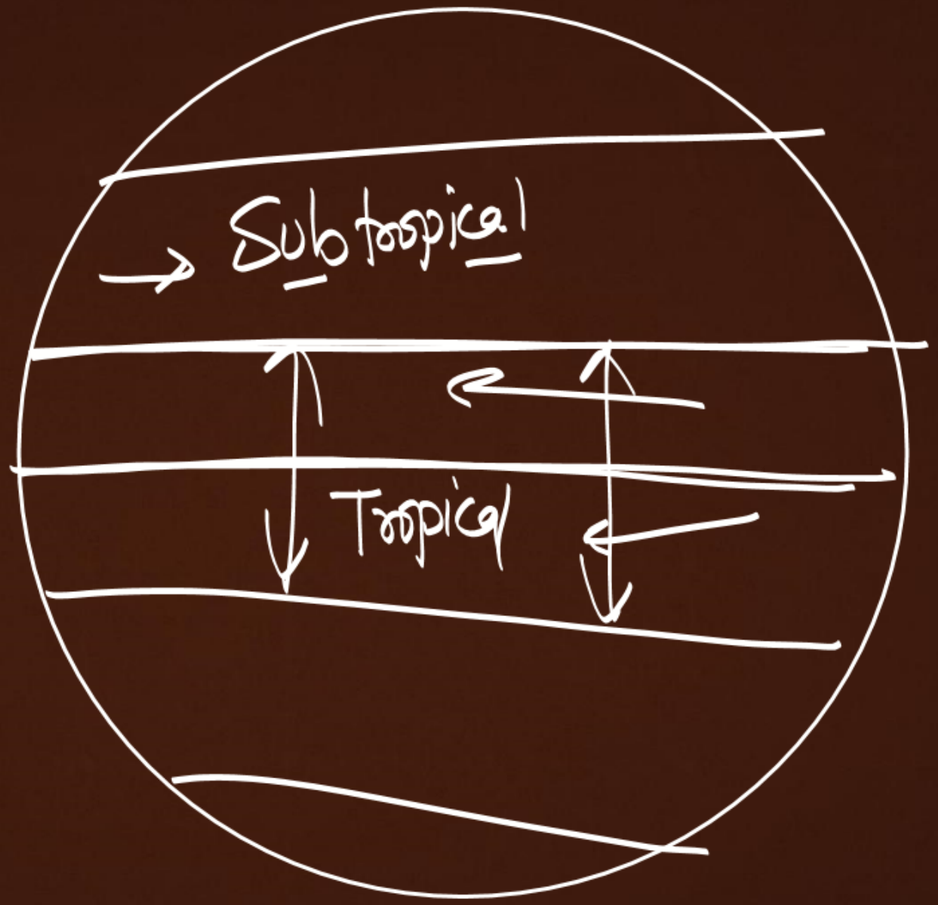
ITCZ

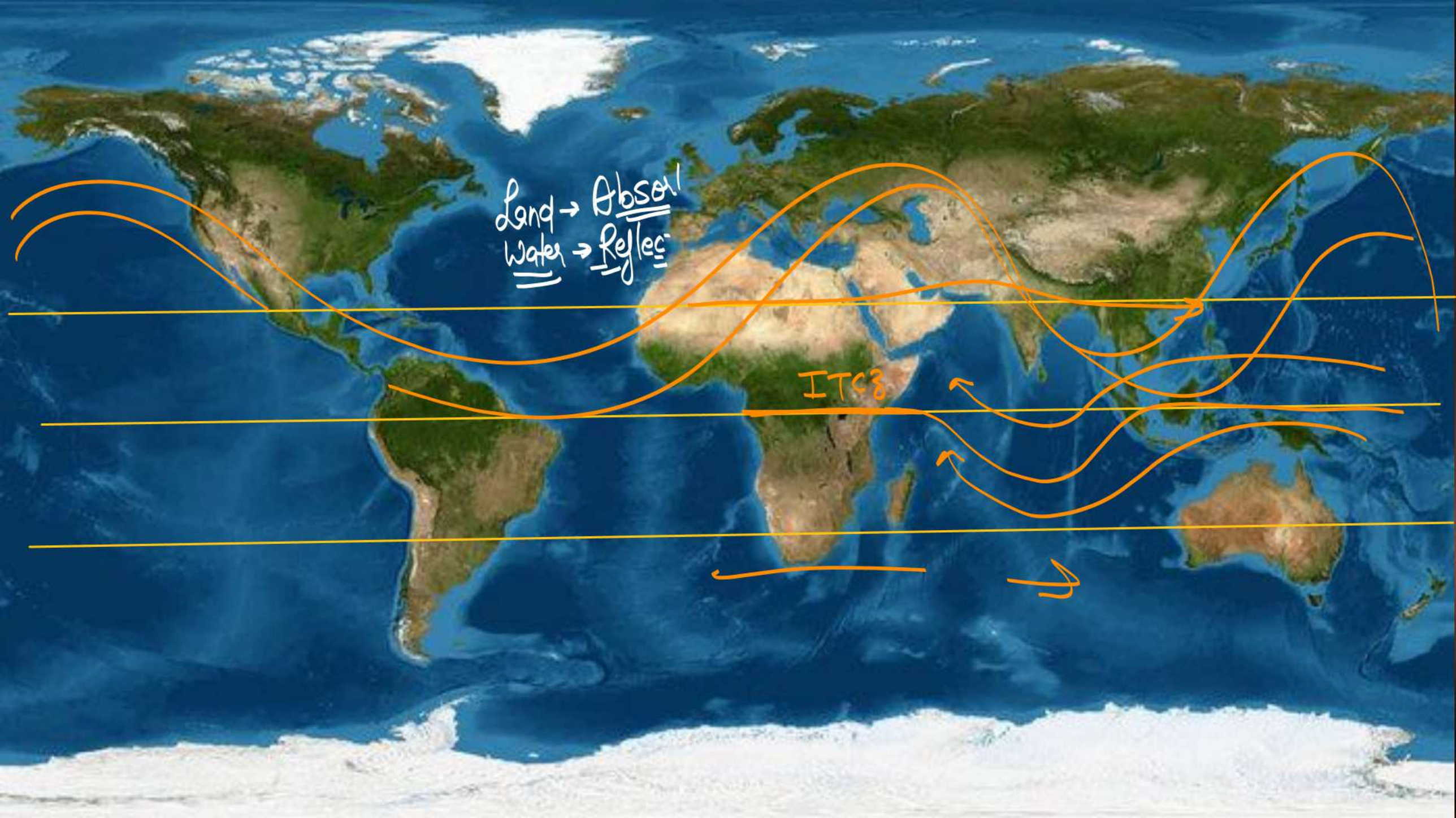
Inter - between

Tropical → Tropics

Convergence - meeting

Zone - Area




Land → Absorb
Water → Reflec
ITCZ

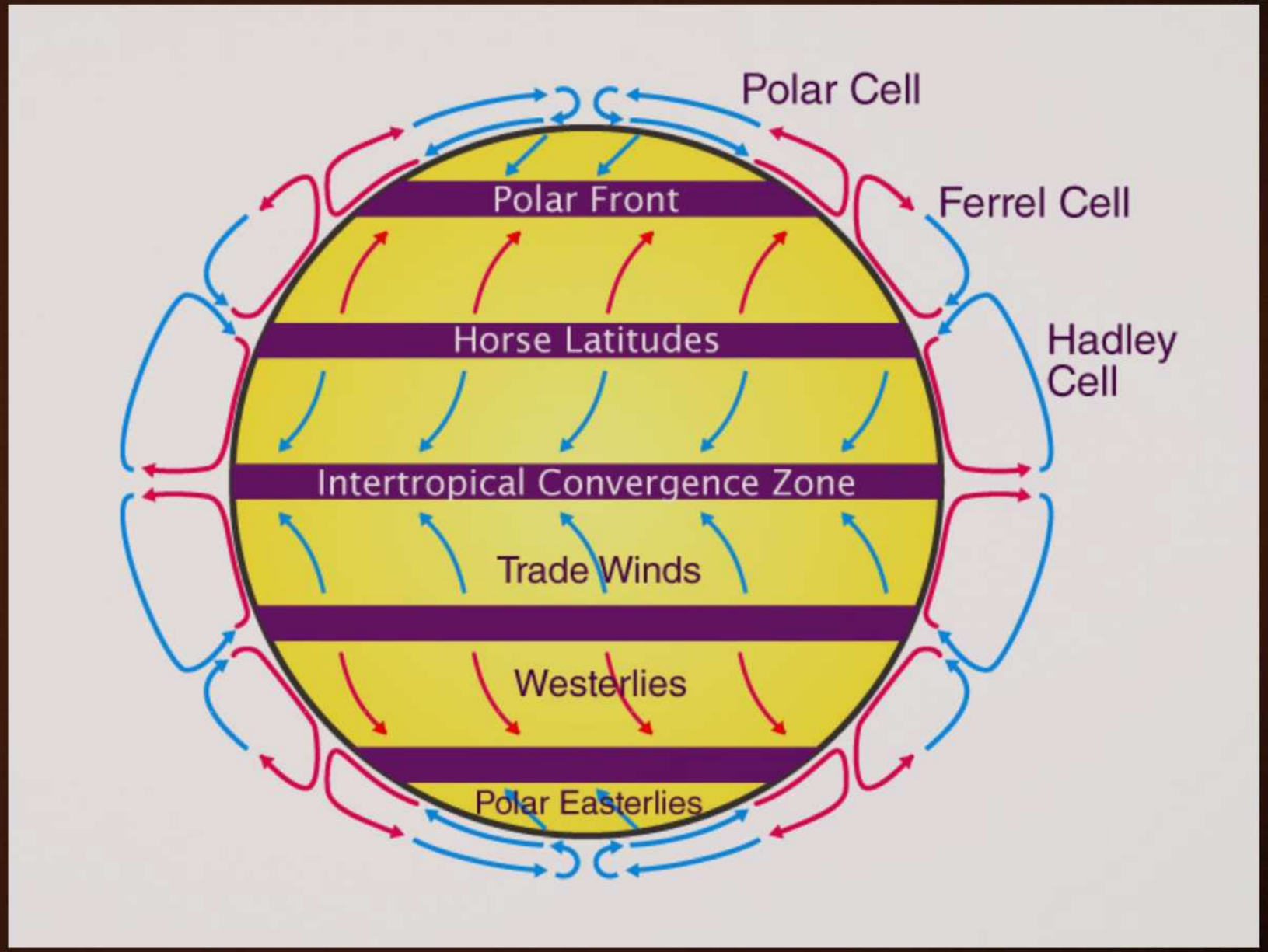
Factors Affecting the Climate of India

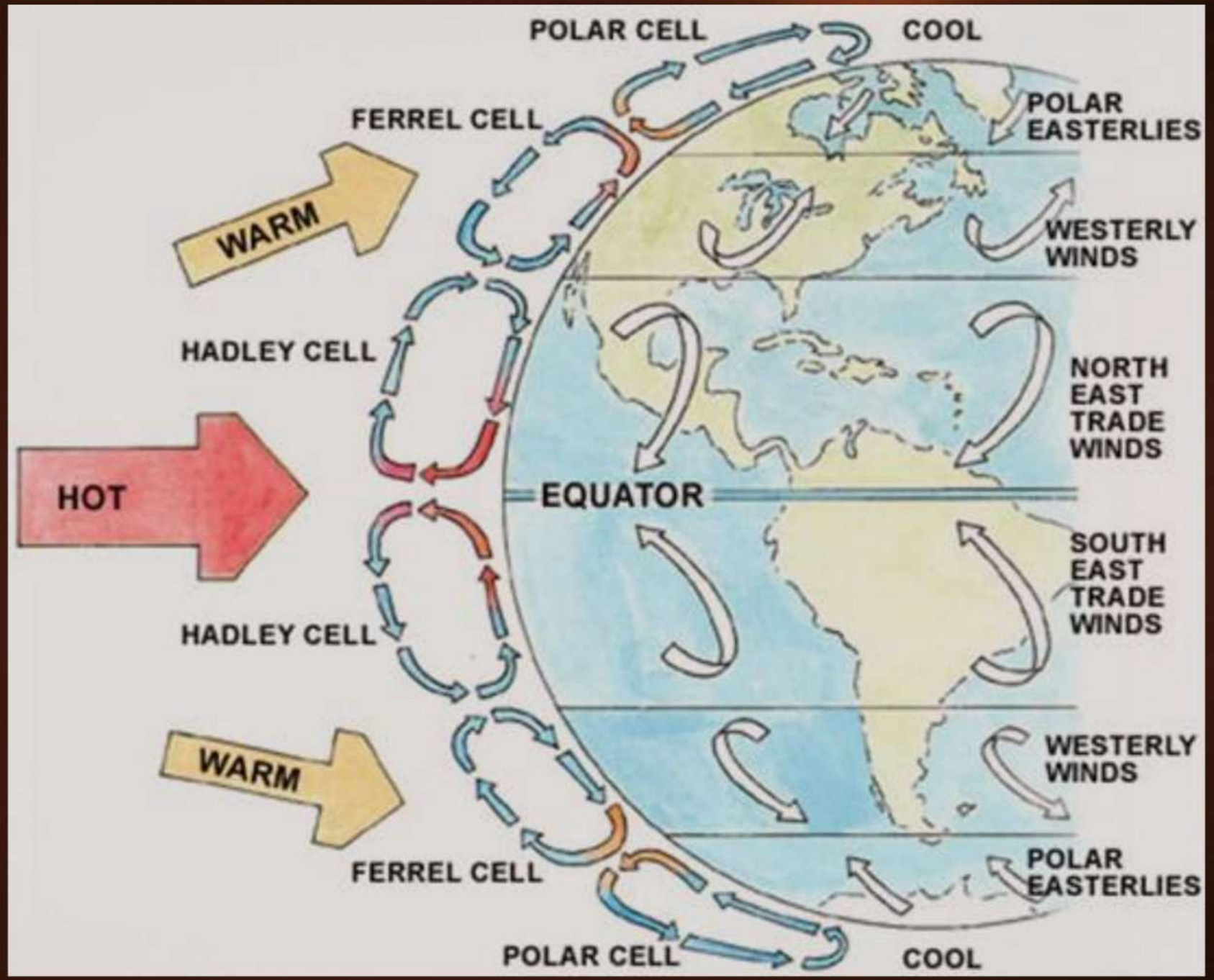
7. **Western Disturbances:** The weather conditions during winter are generally influenced by the distribution pattern of pressure in Central and West Asia.

Summer → Easterly

Winter → N - Westerly
S → Eastery

- In the winter season due to the shifting of the pressure belts, the westerlies blow from 20° N to 50° N Latitude and north India comes under the influence of the winds and cyclones rising from the Mediterranean Sea. These cyclones bring rain to Northern Plains and snow in Jammu and Kashmir in India.

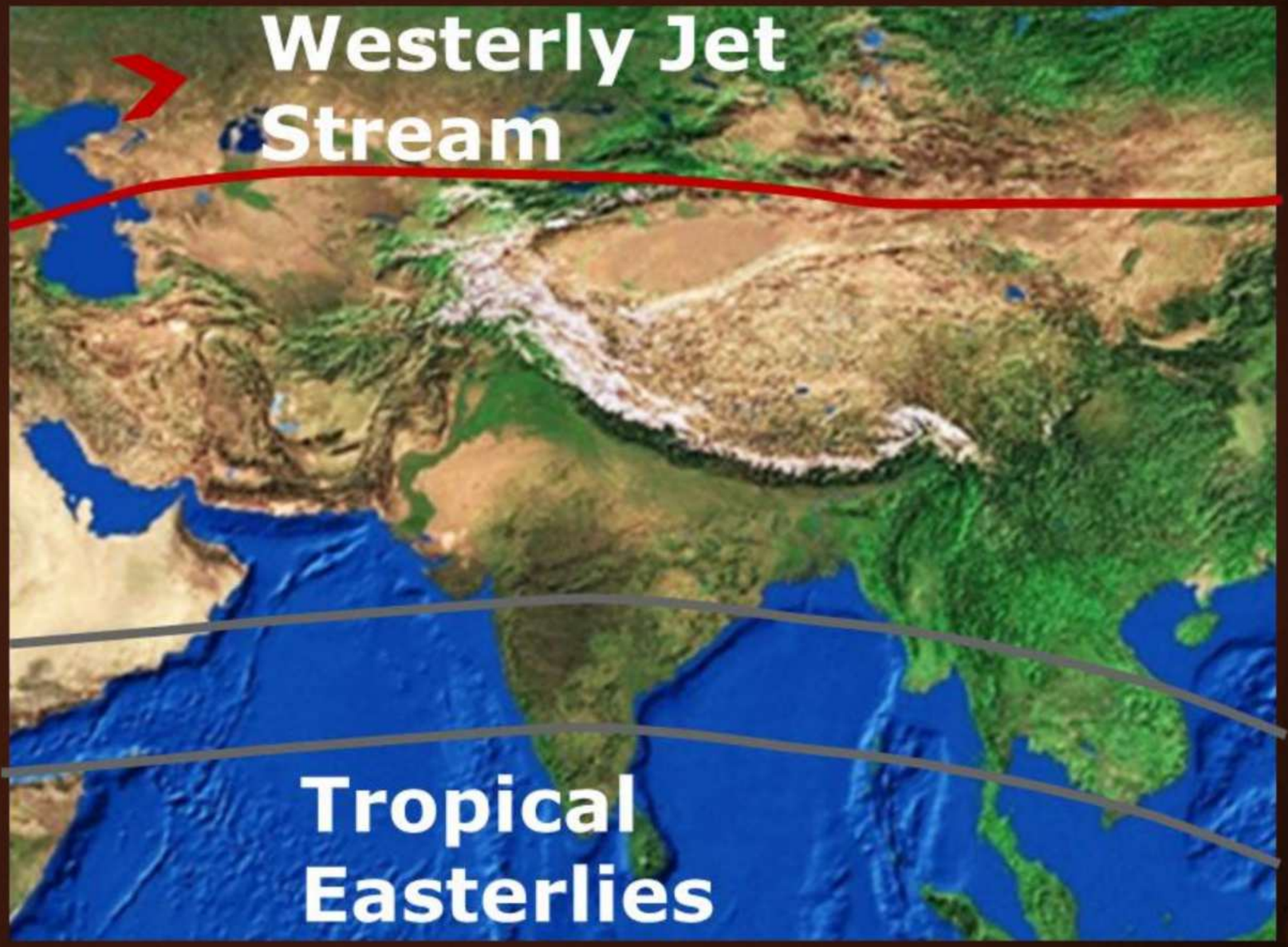

Polar Cell
Ferrel Cell
Hadley Cell
Polar Front
Horse Latitudes
Intertropical Convergence Zone
Trade Winds
Westerlies
Polar Easterlies

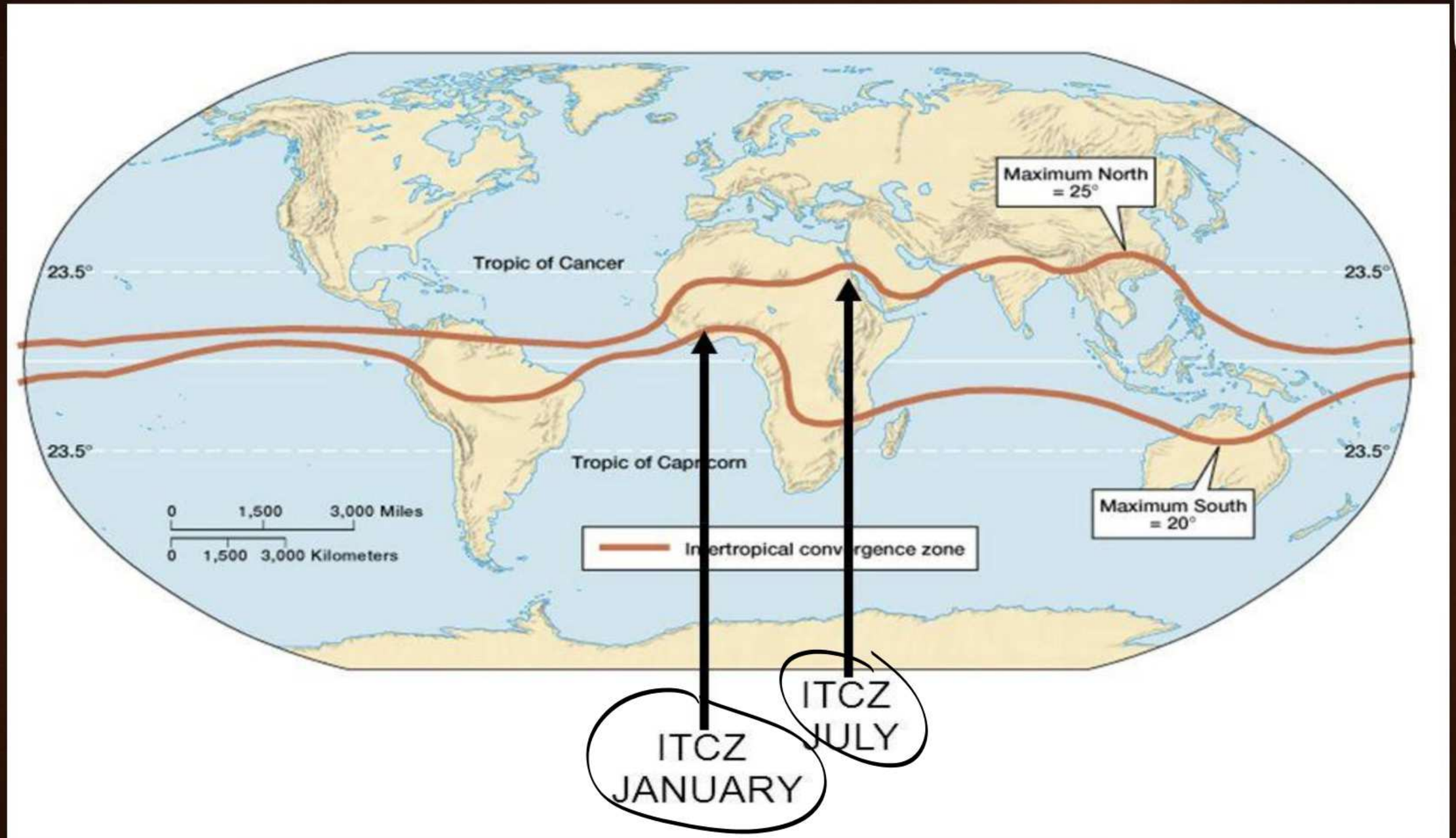

POLAR CELL
COOL
FERREL CELL
POLAR EASTERLIES
WARM
WESTERLY WINDS
HADLEY CELL
NORTH EAST TRADE WINDS
HOT
EQUATOR
SOUTH EAST TRADE WINDS
HADLEY CELL
WESTERLY WINDS
WARM
POLAR EASTERLIES
FERREL CELL
POLAR CELL
COOL

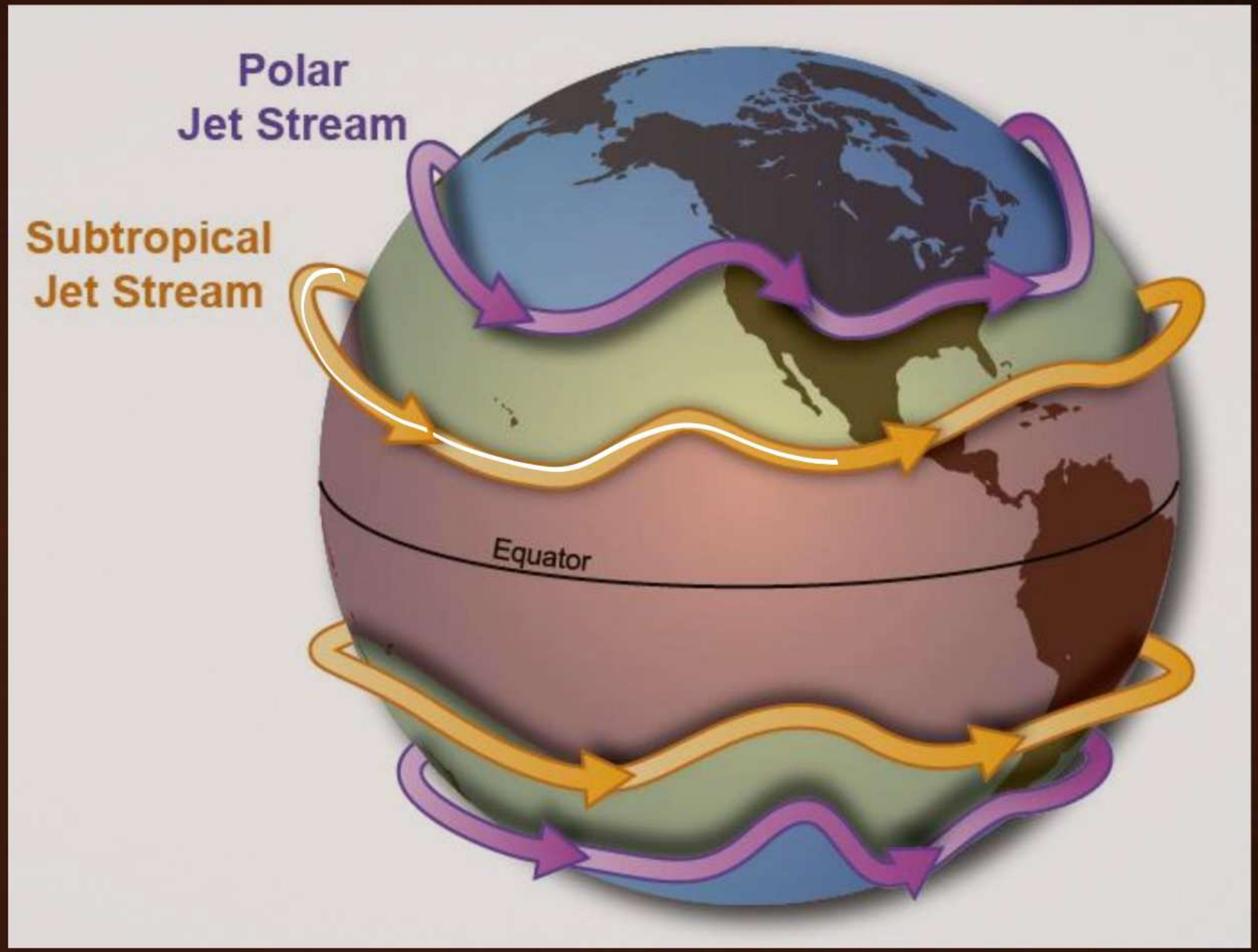
Factors Affecting the Climate of India

8. **Jet Streams:** Jet streams are cold fast blowing winds that develop in the upper layers of the atmosphere. They influence the climate of India.

- The westerly jet stream prevails over the North Indian Plains during the winter months, while the easterly jet stream steers the tropical depression over India.
- These depressions play a significant role in the distribution pattern of the monsoon rainfall in the subcontinent. The highest rainfall occurs along the track of these depressions.


Westerly Jet
Stream
Tropical
Easterlies


Maximum North
= 25°
Tropic of Cancer
23.5°
23.5°
23.5°
23.5°
Tropic of Capricorn
Maximum South
= 20°
0 1,500 3,000 Miles
0 1,500 3,000 Kilometers
Intertropical convergence zone
ITCZ
JULY
ITCZ
JANUARY

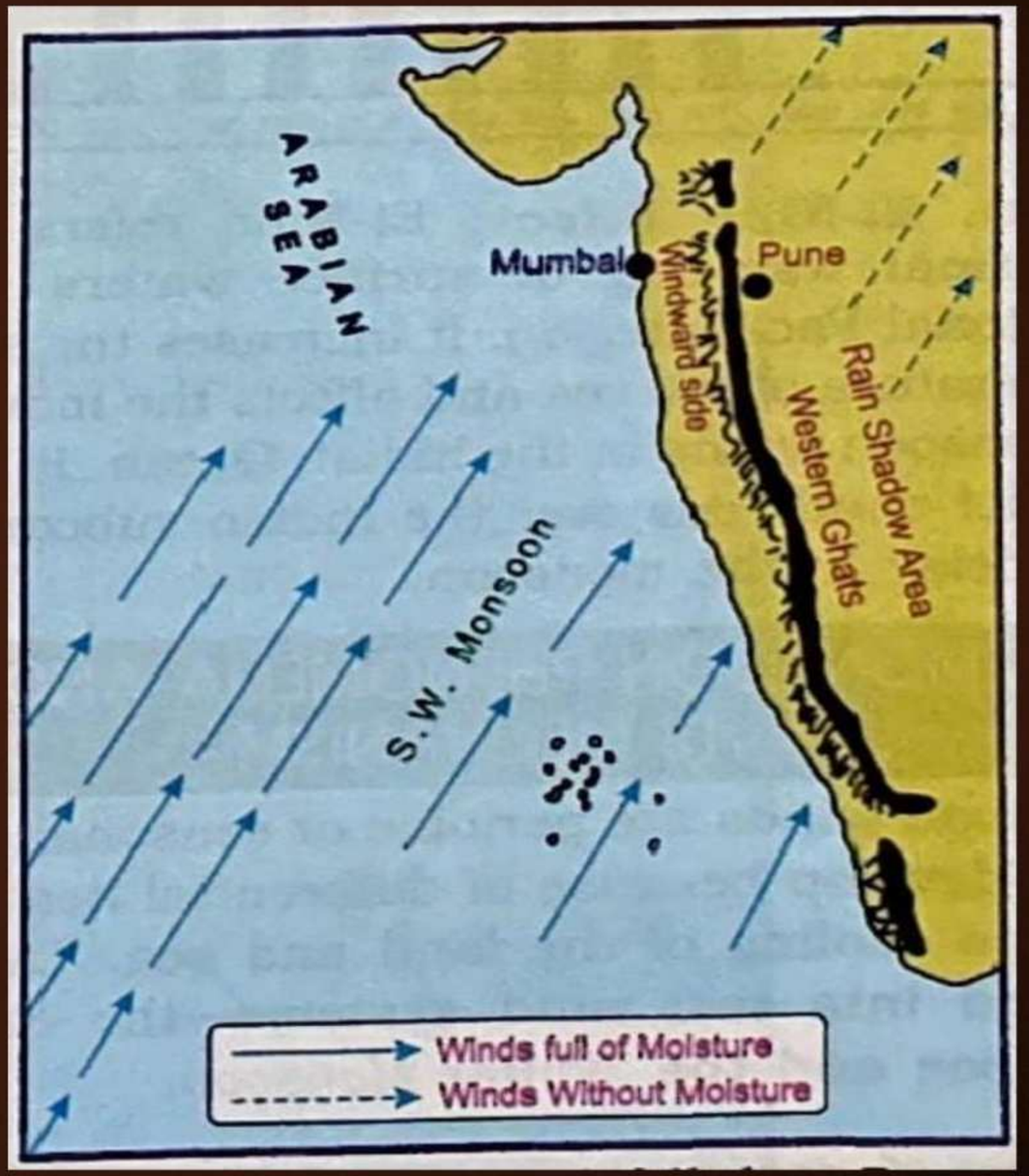

Polar
Jet Stream
Subtropical
Jet Stream
Equator

Factors Affecting the Climate of India

9. **Distance from the Sea:** Areas in the interiors of India have extreme type of climate or continental climate whereas coastal areas have equable or maritime climate. The effect of land breeze and sea breeze caused by differential rate of heating and cooling of land and sea are responsible for moderate climate in coastal areas.
In the Peninsula, the Western Ghats prevent the winds from carrying such benefits far inwards.

Therefore, the moderating influence of the sea is limited to the coastal areas.

Mumbai gets more rainfall than Pune

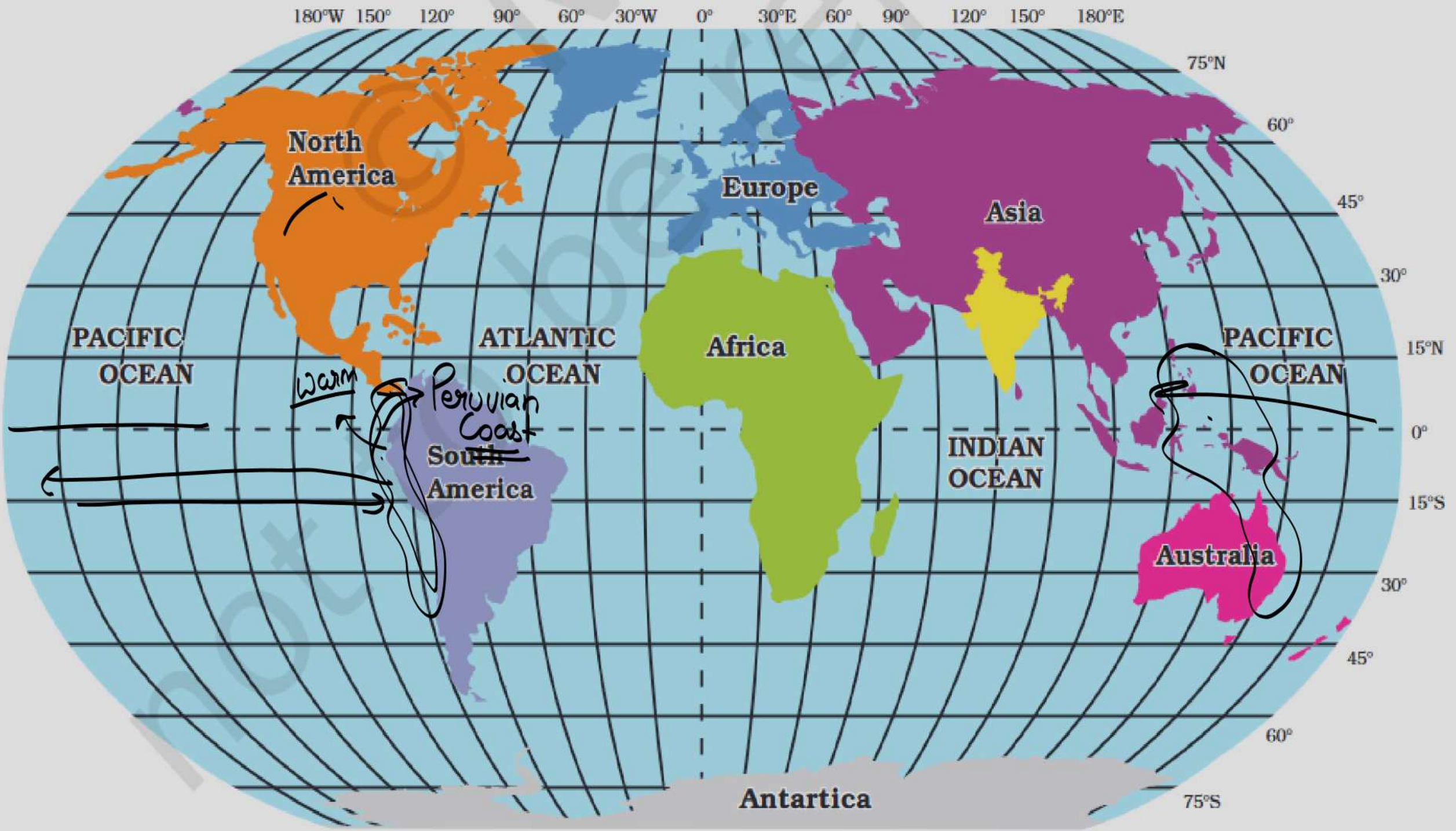


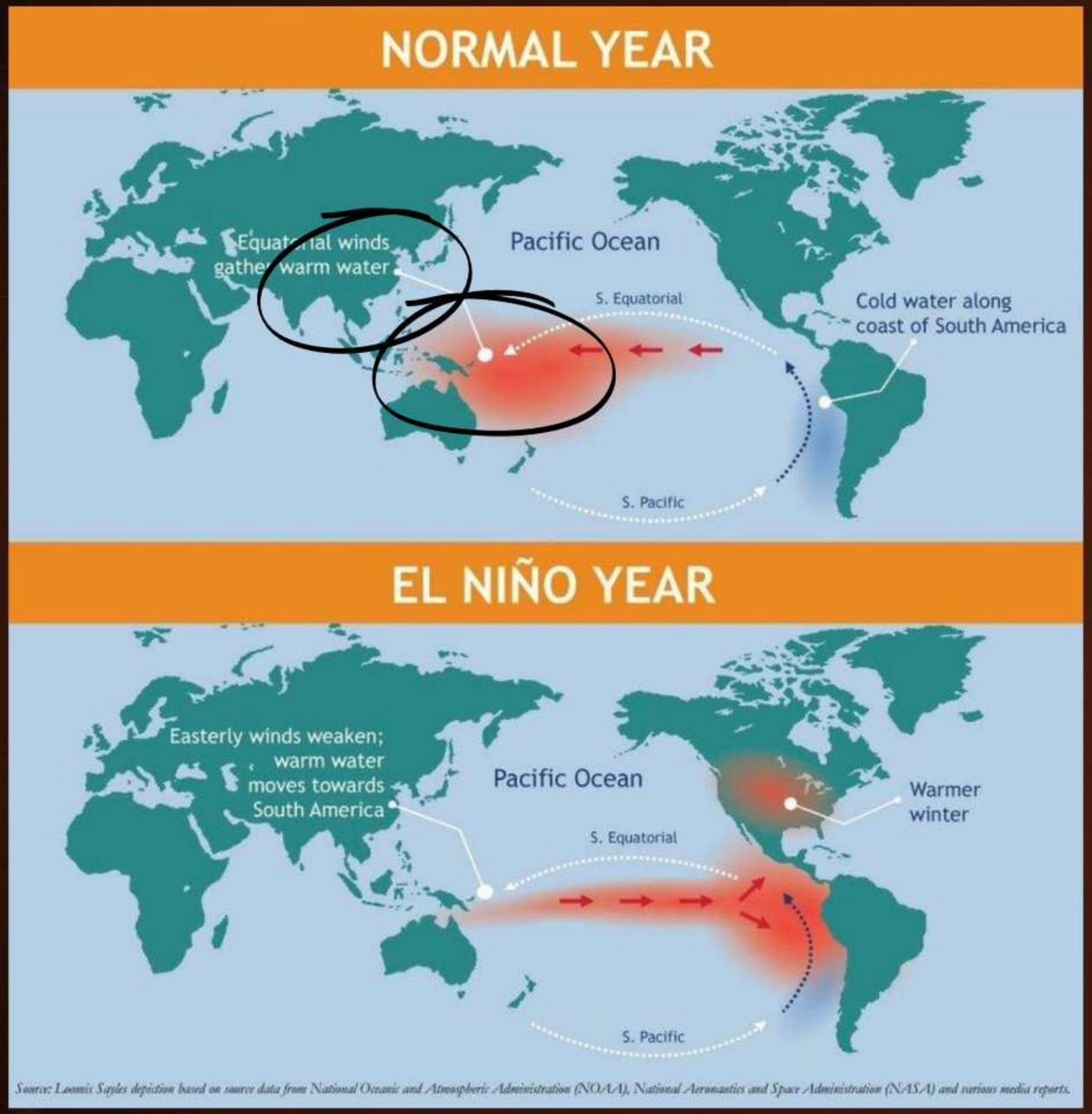

DELHI
Temperature in °C
Rainfall in mm
INDEX
Rainfall (mm)
Average Temp (°C)
JAN FEB MAR APR MAY JUN JUL AUG SEP OCT NOV DEC
CHENNAI
Temperature in °C
Rainfall in mm
INDEX
Rainfall (mm)
Average Temp (°C)
JAN FEB MAR APR MAY JUN JUL AUG SEP OCT NOV DEC
KOZHIKODE
Temperature in °C
Rainfall in mm
INDEX
Rainfall (mm)
Average Temp (°C)
JAN FEB MAR APR MAY JUN JUL AUG SEP OCT NOV DEC
CHERRAPUNJI
Temperature in °C
Rainfall in mm
INDEX
Rainfall (mm)
Average Temp (°C)
JAN FEB MAR APR MAY JUN JUL AUG SEP OCT NOV DEC

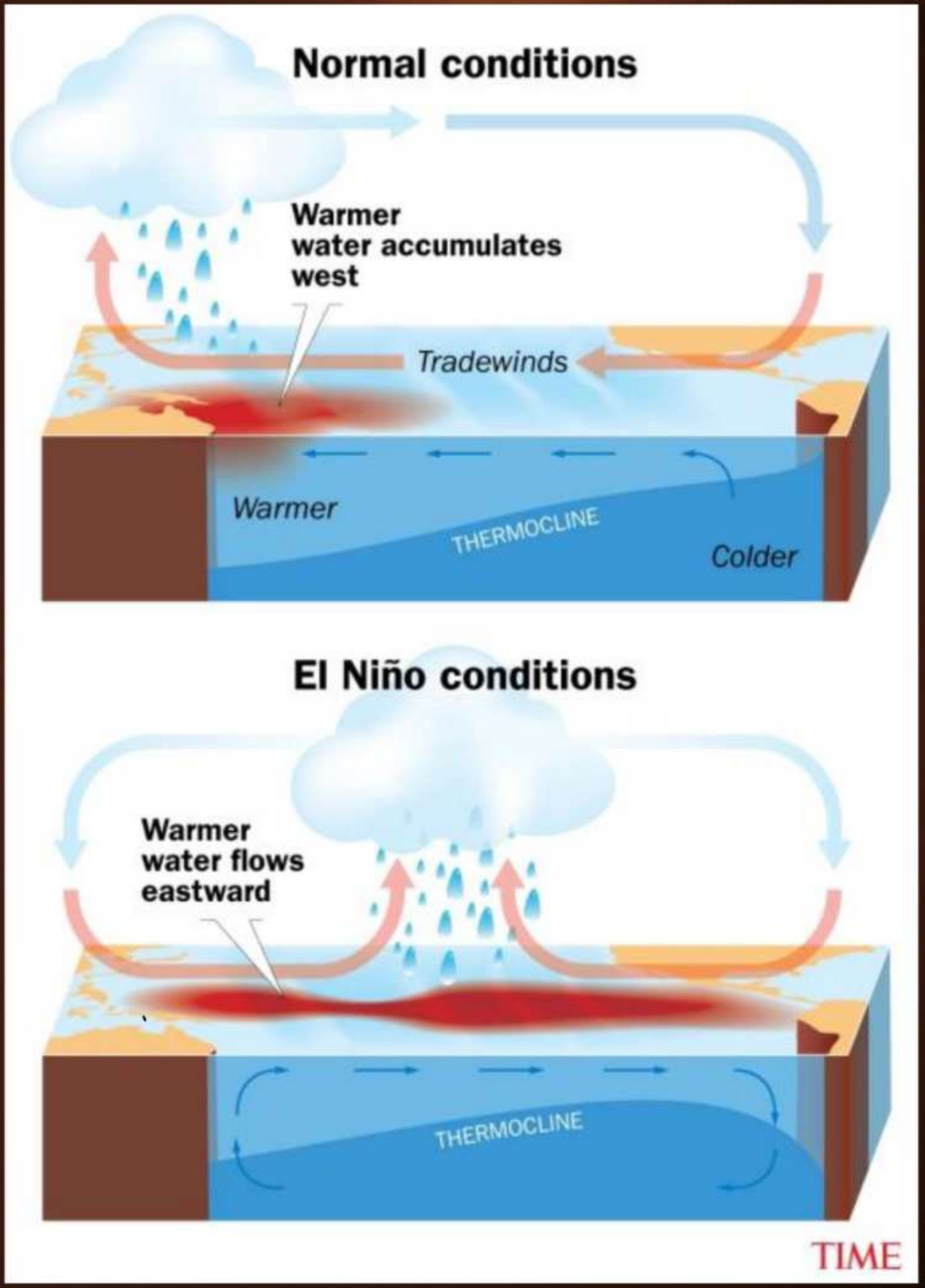
Factors Affecting the Climate of India

10. **El-Nino Effect:** El-Nino refers to an abnormal warming of surface waters in the equatorial Pacific region.

- It increases the surface temperature of the sea and affects the movement of monsoon winds in the Indian Ocean. It results in drier conditions over the Indian subcontinent and delay in the monsoon.


180°W 150° 120° 90° 60° 30°W 0° 30°E 60° 90° 120° 150° 180°E
75°N
60°
45°
30°
15°N
0°
15°S
30°
45°
60°
75°S
North America
Europe
Asia
Africa
PACIFIC OCEAN
ATLANTIC OCEAN
INDIAN OCEAN
PACIFIC OCEAN
South America
Warm
Peruvian Coast
Australia
Antartica


NORMAL YEAR
Equatorial winds gather warm water
Pacific Ocean
S. Equatorial
Cold water along coast of South America
S. Pacific
EL NIÑO YEAR
Easterly winds weaken; warm water moves towards South America
Pacific Ocean
Warmer winter
S. Equatorial
S. Pacific
Source: Loomis Sayles depiction based on source data from National Oceanic and Atmospheric Administration (NOAA), National Aeronautics and Space Administration (NASA) and various media reports.


Normal conditions
Warmer water accumulates west
Tradewinds
Warmer
THERMOCLINE
Colder
El Niño conditions
Warmer water flows eastward
THERMOCLINE
TIME

Climate → India → Tropical monsoon → Mausim → Seasonal reversal of wind.

- → Half Tropical Half-Subtropical.
- → Continentality / Maritime influence → Temperature
- → Relief feature → Precipitation / Rainfall.

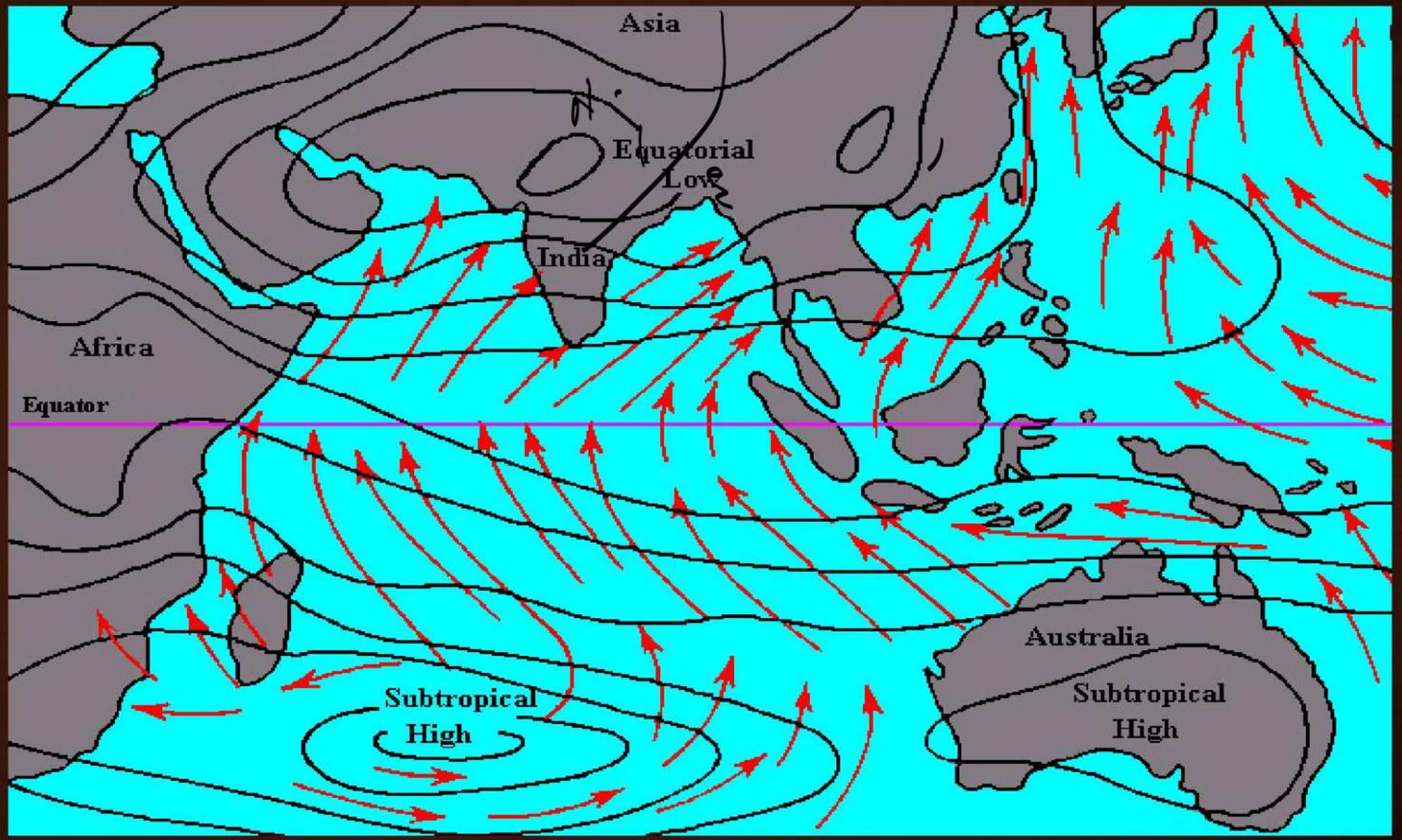
Factors affecting the climate of India.

→ i) Presence of Himalayas.
ii) Relief features → Orographic rain
→ Rain shadow area.
iii) Monsoon → South-Western
→ North-Eastern.
iv) Latitudes
v) Altitude
vi) Influence of the Sea
vii) Western disturbances
viii) Jet streams
ix) El-Nino Effect.

Monsoon Phenomena and its Mechanism



- Monsoon winds are periodic or seasonal winds.
- They develop because of differential heating as well as cooling of the land and sea.
- They are divided into two wind systems-the Summer Monsoon and the Winter Monsoon.


Asia
Equatorial
Low
India
Africa
Equator
Australia
Subtropical
High
Subtropical
High

Monsoon Phenomena and its Mechanism

Summer Monsoons

- In summer, the land gets heated more than the sea.
- Hence, a centre of low pressure develops on the land.
- Over the adjoining sea, the air is comparatively cool, and a high pressure develops there.
- This causes the winds to blow from the sea to the land.
- It is the 'Summer Monsoon.'
- In May, June and July, the plains of the Indian subcontinent are heated by the vertical rays of the sun.

Monsoon Phenomena and its Mechanism

- The intense heat develops a low pressure.
- During these months, a high pressure area develops over the Indian Ocean.
- So, the winds blow from the Indian Ocean northward and north-westward into Asia.
- As they blow from the sea to the land, they bring heavy rainfall in some parts of the Indian subcontinent.
- The summer monsoon winds blow south-west; so they are known as the 'South-West Summer Monsoon.'

Monsoon Phenomena and its Mechanism

Winter Monsoons

- During winter season, the conditions are the reverse of those in summers.
- A high pressure develops over a big landmass stretching from Central Asia up to north-west Indian plains.
- At the same time a low pressure zone develops in the Indian Ocean.
- As the winds blow from the land to the sea, they bring cold dry weather.
- They are incapable of producing rain.

Monsoon Phenomena and its Mechanism



- When these winds blow over seas and pass over the adjoining land, they bring some rainfall.
- The Southern Coromandel Coast (Tamil Nadu and southern tip of Andhra Pradesh) in India get rain from winter monsoons.
- The winter monsoon winds blow north-east; so the monsoon is known as the 'North-East Winter Monsoon.'

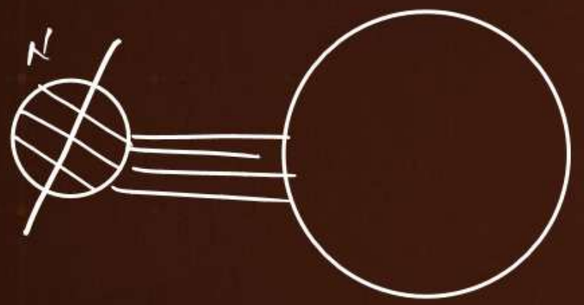
Four Seasons

- The Monsoon type of climate has distinct seasonal pattern marked by significant change from one season to the other.
- These changes are clearly visible in the interior parts of the country.
- The coastal areas, due to the influence of the seas do not experience much variation in temperature though there occurs variation in the pattern of rainfall.
- Thus, according to the India Meteorological Department (IMD) the year may be divided into four principal seasons on the basis of monsoon variations:

Four Seasons

1. The Hot and Dry Summer (March to May).
2. The Hot and Wet or Rainy Season or the South-West Monsoon (June to September). India-Pressure, Prevailing Winds and Cyclones in (April)
3. The Retreating South-West Monsoon or Post-monsoon (October-November).
4. The Cold and Dry Winter Season or NorthEast Monsoon (December-February).

1. THE HOT DRY SUMMER



- In India, the hot season begins in March and lasts until June.
- The vertical rays of the sun fall directly over the Tropic of Cancer during this period.

Temperature

- From March to May, due to the apparent northward movement of the path of the sun's vertical rays, the length of the day increases.
- It results in the increase in solar radiation which moves northward over the tropical areas of South Asia and the temperature starts rising.

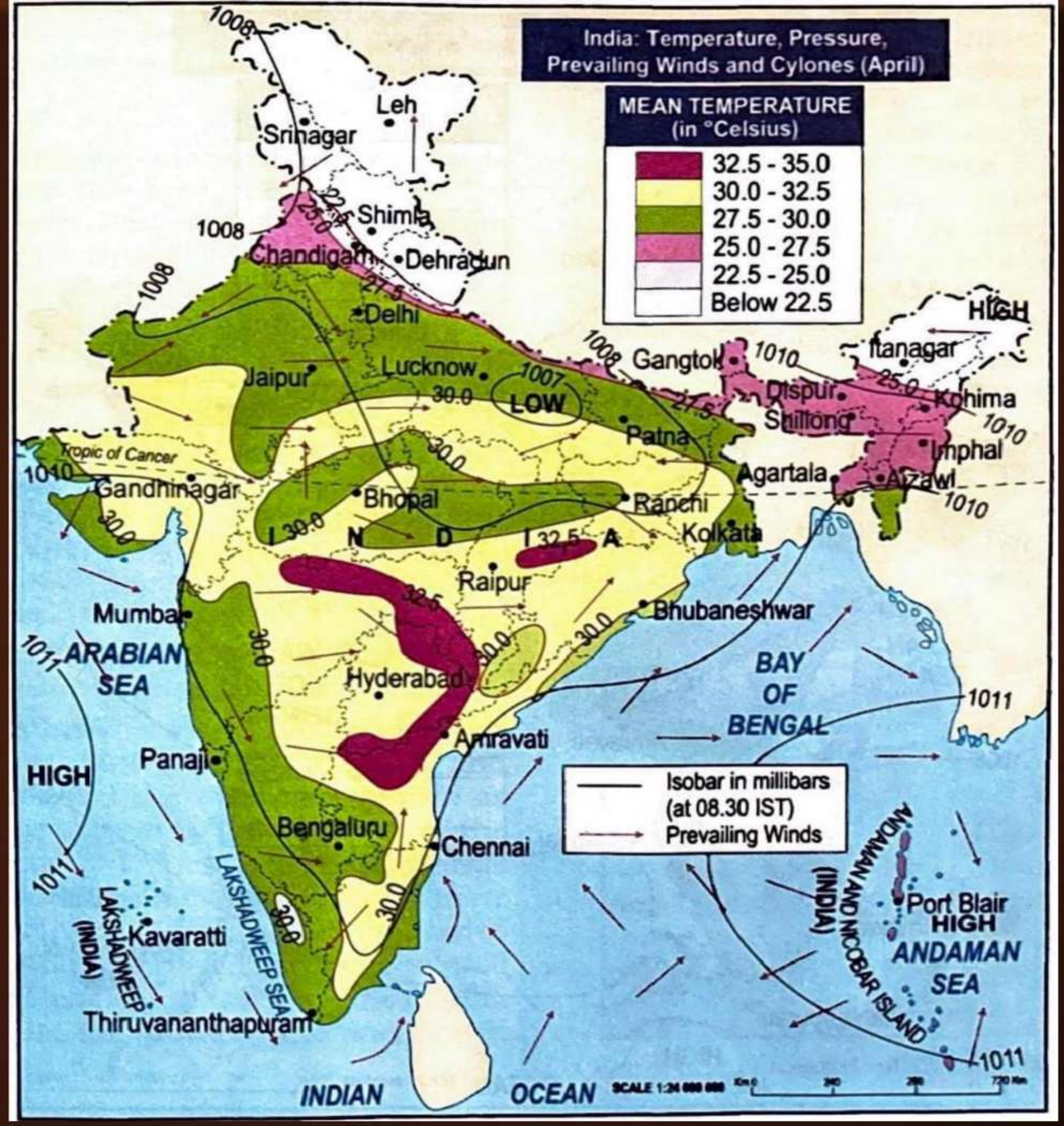
1. THE HOT DRY SUMMER

- In most parts of India, temperature ranges between 30° C and 32° C.
- The highest day temperatures increase as the heat belt moves further north.
- In north-western part of India, temperatures around 48° C are not uncommon.
- In south India, the hot weather is not as intense as in north India.

1. THE HOT DRY SUMMER

- The moderating influence of the oceans together with the Peninsular shape of south India keeps the temperatures lower than that in north India.
- The temperatures, therefore, remain between 26° C and 32° C in south India.
- There is some respite from the heat in the coastal regions due to the influence of the sea.
- Plateaus and hills are also relatively cool because of the elevation.
- The diurnal variation is large, especially in the interior part lying on the west of the subcontinent.

India Mean Temperatures (July)



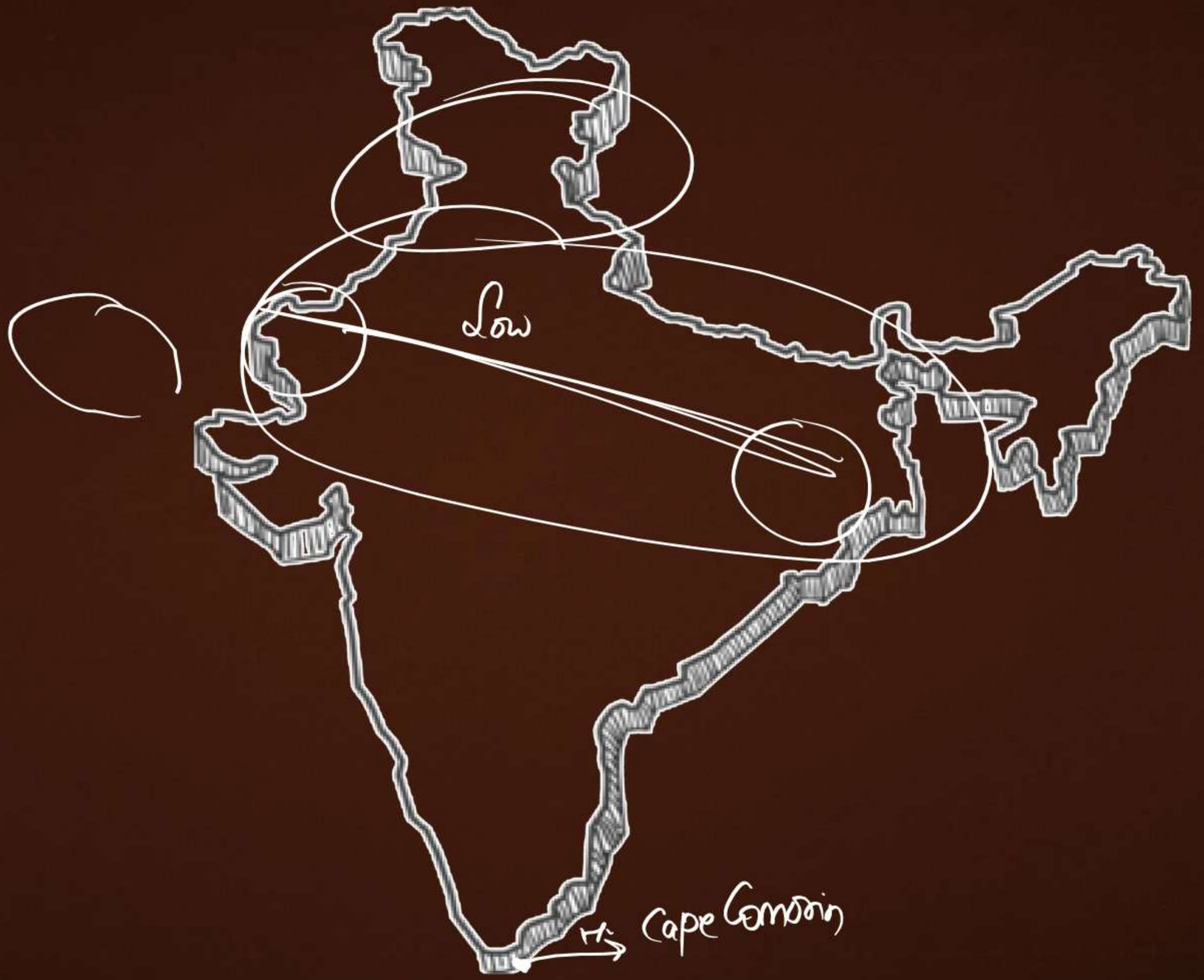
1. THE HOT DRY SUMMER

Pressure Conditions

- The warmest area slowly shifts from the Deccan to north west India.
- The high temperature in the subcontinent causes a low pressure between Thar Desert and Chota Nagpur Plateau.
- The surrounding seas are cooler and develop high pressure conditions in the Indian Ocean.
- By the end of May a comparatively high pressure area develops near Cape Comorin extending towards the Arabian Sea.

1. THE HOT DRY SUMMER

- By June, the inland low pressure system becomes more intense and moves northward with its centre over the Indus lowlands and Baluchistan.
- The influx of these winds by mid-June brings about a change in the weather towards the rainy season.


Low
Cape Comorin

1. THE HOT DRY SUMMER

Storms and Rainfall

- The hot-dry weather is marked by weak winds and dryness over the area.
- The low pressure over the Northern Plains draws winds from the surrounding areas and gives rise to thunderstorms with strong dusty winds.
- These winds cause a shower of rain mostly in Punjab, Haryana and Uttar Pradesh.

1. THE HOT DRY SUMMER

Local Winds

(a) A striking feature of the hot weather season is the strong and dusty winds, called loo, which blow during the day over northern and northwestern India.

These winds have temperature range between 45° C and 50° C which is hot enough to cause heatstrokes.

(b) There are occasional tornado-like dust storms in Punjab and Haryana and Uttar Pradesh.

1. THE HOT DRY SUMMER

(c) The thunderstorms accompanied with strong winds and heavy rainfall occur in Assam and West Bengal.

These local winds are known as Kalbaisakhi which means the calamity of the month of Baisakh'.

In Assam, these storms are known as Bardoli Chheerha.

They originate over the Chota Nagpur Plateau and are carried eastward by westerly winds.

They bring rainfall in Assam, West Bengal and Odisha.

This rainfall is quite beneficial for growing jute and rice in West Bengal and tea in Assam.

1. THE HOT DRY SUMMER

(d) Thunderstorms cause rainfall along the Kerala and Karnataka coasts.

The little rainfall even that they bring is important for mango, tea and coffee plants.

Such rains are called mango showers as they help in the early ripening of the mango crop.

They are also called cherry blossoms in Karnataka.

Thank You